39

Pelvic Floor Disorders

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SUMMARY PEARLS

- Pelvic floor disorders occur in both sexes.
- Musculoskeletal causes of pelvic floor disorders are important to include in the differential diagnosis for symptoms of pelvic pain, urinary or fecal incontinence, urinary urgency and frequency, and defecatory dysfunction.
- Pelvic floor muscle (PFM) dysfunction can be categorized into disorders of increased PFM tone, disorders of PFM pain, disorders of decreased PFM tone, and disorders of PFM coordination.
- Chronic pelvic pain will require different types of treatment interventions compared with acute pelvic pain given the presence of a centralized pain mechanism.
- A thorough history is essential in diagnosing neurogenic pelvic pain caused by the overlap of cutaneous distributions of the border nerves leading to minimal positive exam findings.
- Pelvic girdle pain is a common musculoskeletal complaint found in both pregnant and postpartum populations that responds well to conservative treatment interventions, including physical therapy.

Introduction

Pelvic floor disorders include a wide-ranging group of potentially disabling, embarrassing, and often painful conditions that can greatly affect a person's quality of life. In this chapter, we use the broad term *pelvic floor disorders* to encompass symptoms of pelvic floor muscle (PFM) dysfunction, abnormal urinary or defecatory function, neurogenic pelvic pain, pregnancy and postpartum pelvic disorders, and chronic pelvic pain (CPP). The term *pelvic floor disorders* is a distinct and broader term than *pelvic floor dysfunction* because the latter refers to abnormal functioning of the PFMs in particular, as opposed to the viscera, joints, or nerves. The pelvic floor consists of muscles, fascia, and ligaments that support the pelvic organs and help to provide control for bodily functions. Pathology within the musculoskeletal, visceral, and neurologic structures of the deep pelvis can lead to the development of pelvic pain, dyspareunia, voiding dysfunction (including urinary incontinence [UI] or urinary urgency and frequency), fecal incontinence (FI), and constipation (Table 39.1).

Either sex can develop symptoms of pelvic floor dysfunction, although females are at increased risk because of their unique anatomy and biomechanics. The female pelvis is broader and shallower, requiring greater muscular and ligamentous stiffness to provide support and stability.[143] Females are also more likely to experience physiologic pelvic changes or incur injury to the pelvic floor as a result of pregnancy and childbirth. As a result, abnormal biomechanics of the PFMs may lead to changes in contraction, relaxation, muscle strength, and myofascial pain.

People with pelvic floor disorders benefit from an interdisciplinary rehabilitation approach to improve function and reduce pain. Many patients undergo procedures or surgical interventions for presumed visceral origins before musculoskeletal causes are considered, which delays diagnosis and treatment of musculoskeletal dysfunction, with or without pelvic pain.[3,34,56] Physiatrists with experience in acute and chronic pain, neurologic and musculoskeletal conditions, and neurogenic bowel and bladder management are well suited to direct such a patient's care.[143] In addition to diagnosing and managing the patient's pelvic floor disorder medically, the physiatrist plays a key role in providing a detailed prescription for physical therapy (PT), including for specialized pelvic floor physical therapy (PFPT). This prescription allows the physiatrist to convey impressions and suggest specific interventions for the pelvic floor as well as other related musculoskeletal structures (e.g., lumbar spine, pelvis, and hip). The rehabilitation provider must be aware of when to consult obstetrics-gynecology, urogynecology, urology, colorectal surgery, gastroenterology, psychiatry, and/or psychology colleagues to provide additional specialist care. This chapter reviews the anatomy and physical examination of the pelvic floor, discusses the definitions and epidemiology of pelvic floor disorders, and explains the rehabilitation approach to treating these common conditions.

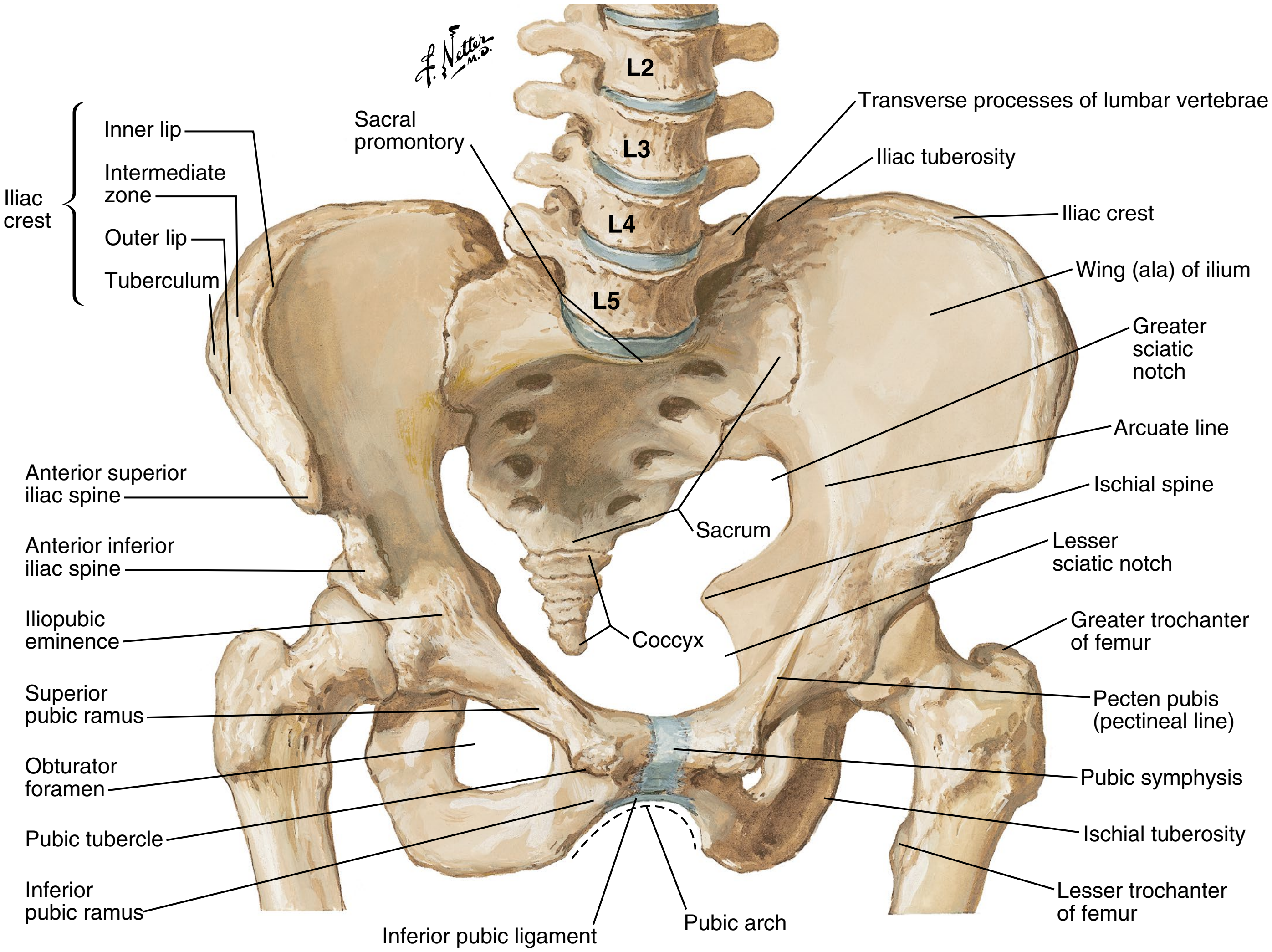
Pelvic Floor Neuromuscular Anatomy

The pelvic floor is composed of muscles, ligaments, and fascia that act as a sling to support the bladder, reproductive organs, and rectum. This sling of soft tissue is enclosed by the bony scaffolding of the pelvis, formed from the ilium, ischium, and pubis; the latter articulate with the sacrum posteriorly and each other anteriorly (Fig. 39.1). Extending from the sacrum is the coccyx, which acts as an important ligamentous and tendinous anchor.

TABLE 39.1 Possible Etiologies of Pelvic Floor Disorders

Urologic	Gynecologic	Gastrointestinal	Dermatologic	Neuromuscular	Psychological
Interstitial cystitis/painful bladder syndrome Chronic prostatitis	Endometriosis Vulvodynia Pelvic congestion syndrome Pelvic organ prolapse Genitourinary syndrome menopause	Irritable bowel syndrome	Lichen sclerosus Lichen planus	Pelvic floor muscle dysfunction Sacroiliac joint pain/dysfunction Pubic symphyseal dysfunction Peripheral neuropathies (border, pudendal, etc.)	Anxiety disorders Physical, sexual, or emotional abuse

Adapted from Le PU, Fitzgerald CM. Pelvic pain: an overview. *Phys Med Rehabil Clin N Am*. 2017;28(3):449-454; and Carey ET, Moore K. Updates in the approach to chronic pelvic pain: what the treating gynecologist should know. *Clin Obstet Gynecol*. 2019;62(4):666-676.



• **Fig. 39.1** Bony framework of pelvis. (Courtesy www.netterimages.com.)

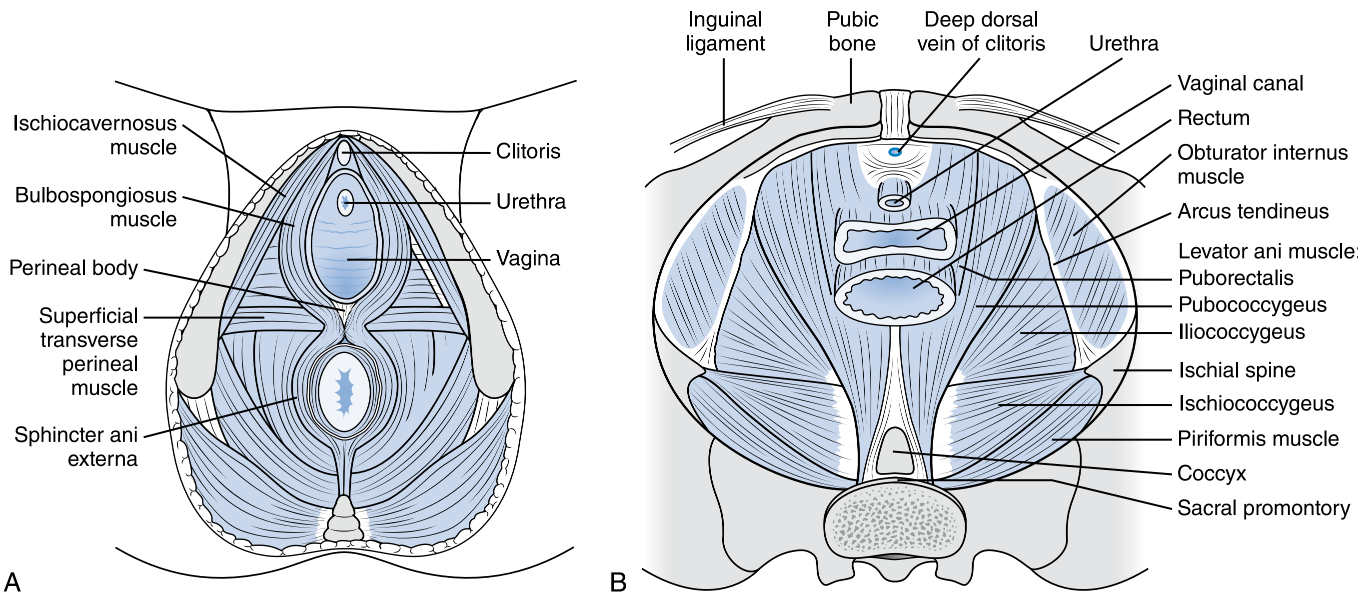
The stability of the articulating surfaces of the posterior pelvis is thought to arise through the mechanisms of force closure and form closure. Form closure is achieved through the interlocking of the ridges and grooves of the bony joint surfaces in the pelvis, whereas force closure is achieved through the compressive forces of the muscles, ligaments, and fascia, providing stability.[141,187] In the posterior pelvic ring are two sacroiliac joints (SIJs). The anterior sacroiliac ligaments—comprising the anterior longitudinal ligament, the anterior sacroiliac ligament, and the sacrospinous ligament—stabilize the joint by resisting upward movement of the sacrum and lateral movement of the ilium. The posterior sacroiliac ligaments are made up of the short and long dorsal sacroiliac ligaments, the supraspinous ligament, the iliolumbar ligament, and the sacrotuberous ligament. These ligaments function to resist downward and upward movement of the sacrum and medial motion of the ilium. Anteriorly, the pubic symphysis functions as a cartilaginous joint between the two pubic bones; it is reinforced by superior, inferior, anterior, and posterior ligaments. Functionally, it resists tension, shearing, and compression and is subject to great mechanical stress as it widens during pregnancy.

The PFMs are divided into two layers: the superficial perineal layer and the deep pelvic diaphragm. The superficial PFMs are the bulbospongiosus, ischiocavernosus, and transverse perineal muscles. The deep PFMs that line the inner walls of the pelvis are the levator ani and coccygeus, which—along with the endopelvic fascia—comprise the pelvic diaphragm (Table 39.2). The levator ani is composed of three muscles: the puborectalis, pubococcygeus, and iliococcygeus (Fig. 39.2). The perineal body or central perineal tendon is located between the vagina and anus. This is a site where the pelvic muscles and sphincters converge to provide support to the pelvic floor. Rupture of this entity during childbirth can lead to pelvic organ prolapse (POP). The PFMs function to support the pelvic organs by coordinated contraction and relaxation.[125] At rest, the pelvic floor provides active support through muscular activity and passive support from the surrounding connective tissue and fascia. Lining the posterior and lateral walls of the pelvis are the piriformis and obturator internus muscles, respectively.

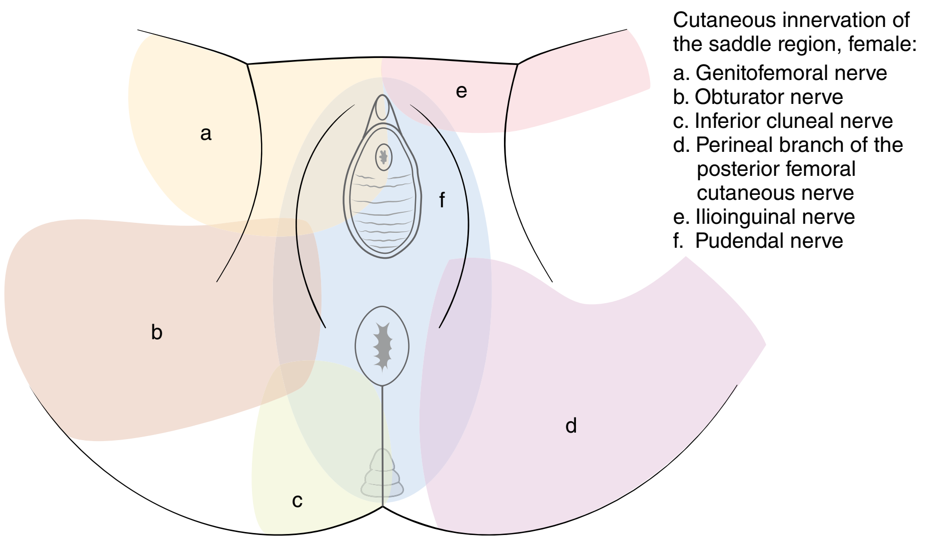
The PFMs receive innervation through somatic, visceral, and central pathways. Cutaneous innervation of the lower abdomen, perineum, and proximal thigh is mediated through the iliohypogastric, ilioinguinal, and genitofemoral nerves (L1-L3). The pudendal nerve (ventral branches of S2-S4) innervates the penis/clitoris, the bulbospongiosus and ischiocavernosus muscles, portions of the levator ani muscles, the perineum, the anus, the external anal sphincter, and the urethral sphincter (Fig. 39.3). This nerve contributes to external genital sensation, continence, orgasm, and ejaculation. The pudendal nerve passes between the

TABLE 39.2 Anatomic Origins, Insertions, Innervations, and Functions of the Deep Pelvic Floor Musculature

Muscle	Origin	Insertion	Primary Innervation	Function
Puborectalis	Pubic symphysis	Pubic symphysis	S3-S5, direct innervation from sacral nerve roots	Raises the pelvic floor
Pubococcygeus	Posterior pubic bone and arcus tendineus	Anococcygeal ligament and coccyx	S3-S5, direct innervation from sacral nerve roots	Maintains floor tone in upright position
Iliococcygeus	Ischial spine and arcus tendineus	Anococcygeal raphe and coccyx	S3-S5, direct innervation from sacral nerve roots	Voluntary control of urination
Coccygeus	Ischial spine	Lower sacral and upper coccygeal bones	S3-S5, direct innervation from sacral nerve roots	Support of fetal head
Obturator internus	Pelvic surface of ilium, ischium, and obturator membrane	Posterior surface of greater trochanter	L5, S1-S2 via nerve to obturator internus	Lateral rotator of thigh



• **Fig. 39.2** The muscles of the superficial pelvic floor (A) and deep pelvic floor (B). (Redrawn from Prather H, Dugan S, Fitzgerald C, et al. Review of anatomy, evaluation, and treatment of musculoskeletal pelvic floor pain in women. *PM&R.* 2009;1:346-358.)



• **Fig. 39.3** Cutaneous innervation of the saddle region, female. (Redrawn from Trescot AM, ed. *Peripheral Nerve Entrapments: Clinical Diagnosis and Management*. Springer; 2016.)

piriformis and coccygeus muscle as it traverses through the greater sciatic foramen, over the spine of the ischium, and back into the pelvis through the lesser sciatic foramen. It courses along the lateral wall of the ischiorectal fossa, where it is contained in a sheath of the obturator fascia termed the *pudendal* (or *Alcock*) *canal*.[10] There are three main terminal branches of the pudendal nerve: the inferior rectal nerve, the perineal nerve, and the dorsal nerve of the penis/clitoris. Muscles of the levator ani are also thought to have direct innervation from sacral nerve roots S3-S5.[18] The inferior cluneal nerve (S2-S3) branches off from the posterior femoral cutaneous nerve and innervates the inferior gluteal, ischial, and inferior perineal cutaneous regions, with no motor innervation.

Overview of Terminology

The International Urogynecological Association and the International Continence Society (ICS) established standardized terminology for PFM function and dysfunction,[86,125] which were most recently updated with significant revisions to the nomenclature in 2021 by the ICS.[73] The PFMs function by coordinated contraction and relaxation as a unit. *Voluntary contraction* occurs when the patient contracts the PFMs on demand, historically referred to as a Kegel contraction. *Relaxation postcontraction* occurs when the patient relaxes the PFMs on demand after a voluntary contraction. *Involuntary contraction* of the PFMs occurs preceding a sudden rise in intraabdominal pressure to prevent incontinence, as during a cough. When PFM coordination is normal, perineal descent and pelvic floor relaxation are expected to occur during bear down or replication of the motion of normal urination or defecation. For the remainder of this chapter, for the sake of simplicity and to align with practical usage in the clinical setting, we will refer to the ICS concept of coordination with voiding or defecation as *reverse Kegel*. Contraction and relaxation can be observed during the pelvic floor physical examination, as described later in this chapter.

Based on examination of the pelvic floor, prior nomenclature described four PFM conditions: normal, overactive, underactive, and nonfunctioning PFM. These terms have been supplanted in the latest ICS terminology guidelines[73] with the following categories of pelvic floor diagnoses, all of which fall under the broader umbrella of PFM dysfunction: (1) disorders of increased PFM tone, (2) disorders of PFM pain, (3) disorders of decreased PFM tone, (4) disorders of PFM coordination (including PFM dyssynergia, vaginismus, and anismus), and (5) pudendal neuralgia (which will be treated separately in this chapter in the section on Neurogenic Causes of Pelvic Pain).

In line with the latest ICS terminology guidelines,[73] PFM tone is described as the resting state of tension of the muscle, as determined by the degree of resistance to passive movement upon palpation, and is believed to comprise a contractile component (muscle contraction) and a noncontractile viscoelastic component (stiffness of connective tissue independent of muscle activity). Dyssynergia of the PFMs refers to "a dysfunction of coordination between the PFM and a functional activity, such as PFM contraction when relaxation is functionally required," as with urination or defecation.[73] Dyssynergia of the PFMs, along with the term disorders of increased PFM tone, has essentially replaced the prior term of overactive PFM.[73] The term *underactive PFM* is no longer favored because decreased tone may or may not be related to muscle contractile activity. The term *nonfunctioning PFM* has similarly been discarded in favor of recognizing the underlying reason for lack of PFM movement, which can be an issue of impaired contractile strength, impaired neurologic function, noncontractile viscoelastic stiffening (as in postradiation fibrosis), or because of the patient having difficulty following cues to move the muscles.

Physical Examination of the Pelvic Floor

A thorough neurologic and musculoskeletal examination of the lumbar spine, hips, abdomen, pelvic girdle, lower limbs, and PFMs will guide the differential diagnosis. The pelvic floor examination consists of a vaginal and rectal examination of PFM function and a neurologic examination of the sacral dermatomes and peripheral nerve distributions of the pelvis. A musculoskeletal pelvic floor examination does not exclude the need for gynecologic, urologic, or colorectal evaluation because visceral structures are not typically evaluated. Consent from the patient is required (verbal or [ideally] written) and must be accompanied by a clear explanation of the purpose and components of the exam as well as disclosure of the risk of discomfort (physical and/or psychological). The examination should occur in a private examination room, in the presence of a chaperone who can provide assistance while also serving as a third-party witness for the benefit of both the patient and the examining provider. The examiner must take care to perform the examination in a professional manner with clear, neutral communication given the sensitive nature of this body region and high likelihood for patients to feel anxious or vulnerable. This is particularly salient in the CPP population, in which history of trauma or abuse and psychiatric comorbidities are highly prevalent.[102] Before the start of the examination, the patient should be advised on the right to terminate the examination at any point. Each aspect of the exam should be announced and explained before proceeding, ensuring that there is ongoing consent for each portion of the examination.

The following sections break down the physical examination into categories of inspection, neurologic testing, palpation, and internal functional examination, However, this does not necessarily reflect the order in which examination components must be performed—that is, in practice, portions of the exam are grouped together based on patient positioning and convenience. The examination should always be tailored to the specific presenting symptoms and differential diagnoses being considered.

Inspection

The musculoskeletal pelvic examination can begin with external inspection, assessing for musculoskeletal asymmetries, such as pelvic obliquity or Trendelenburg sign, and noting any surgical incisions, hernias, swelling, cysts, scars, or lesions that may necessitate appropriate referral to another specialist, such as general surgery, gynecology, or dermatology.

In addition, the examiner should visualize the movements of the PFMs. The examiner can observe the lift of the perineal body by cueing the patient to perform a voluntary contraction and can then visualize normal descent of the perineal body with relaxation postcontraction. One can observe the reverse Kegel motion by cueing the patient to replicate the urination or defecation motion. To observe involuntary contraction, the patient can be cued to cough, with which reflexive lift of the perineal body should be seen. If applicable, the vaginal vestibule can then be evaluated for any visible organ prolapse while the patient performs a Valsalva maneuver.

Neurologic Testing

Neurologic testing should begin with assessment of the strength, sensation, and reflexes of the lower extremities (L2-S1 spinal levels). The examiner may then perform an external sensory examination of the S2-S5 sacral dermatomes in the posterior thighs and buttock region, as well as the pelvic peripheral nerve distributions encompassing the groin, labia/scrotum, and perineum (see Fig. 39.3). An anal wink reflex is obtained with light touch near the anus to test the sacral reflex arc. (Note: This may be difficult to elicit in the presence of overactive PFMs with guarding and is often absent in those who have given birth vaginally.) Bulbocavernosus reflex can also be assessed, and assessment of baseline tone, presence of deep anal sensation, and presence of volitional contraction on rectal examination provide further diagnostic information when neurologic dysfunction is suspected (see Chapter 50). The Q-tip test for provoked vulvodynia is performed by lightly touching a cotton swab at vulvar and vestibular sites to elicit any pain or allodynia.

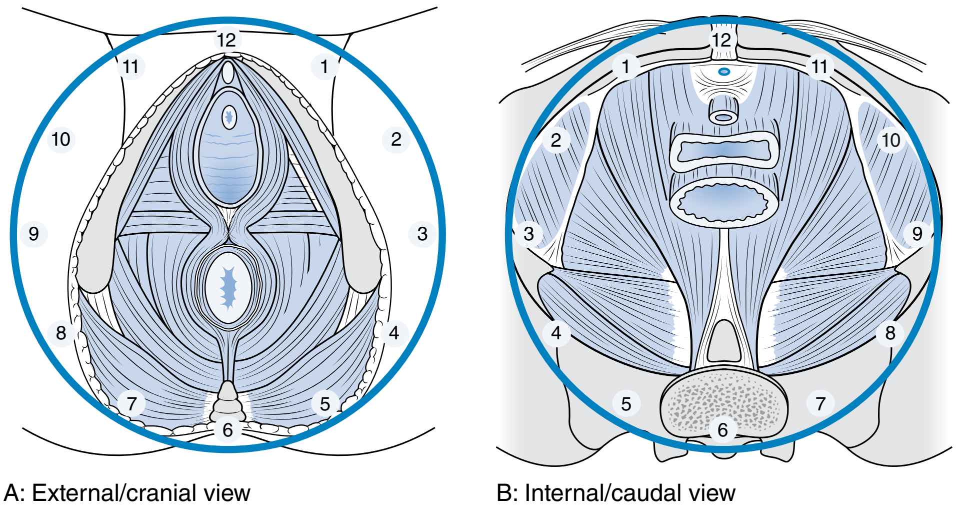
Palpation

The palpatory examination should include identification and palpation of the lumbar spine, SIJs, and pubic symphysis, as well as lumbar, gluteal, and abdominal musculature, assessing for tone and tenderness. The abdominal wall can further be assessed by palpating for rectus diastasis. The superficial PFMs should be palpated for any tenderness, scar, or abnormal tone.

On examination of the internal pelvic floor, both vaginal and/or rectal examinations may be performed depending on the presenting symptoms. It is best to use a flat examining table without stirrups. The vaginal examination is performed in hook lying position, supine with the knees bent, feet planted, and ankles hip-width apart, with the examiner seated near the feet or standing with the table raised. The rectal examination is typically performed in the lateral decubitus position. Before initiating internal examination, we recommend demonstrating the palpation pressure on a neutral area such as the leg and reassuring the patient that internal palpation should not cause significant pain. Remind the patient about the right to terminate the examination at any point if there is excessive pain or discomfort. A lubricated gloved index finger is inserted into the vaginal introitus or anal canal and crooked to palpate the PFMs internally, rotating the wrist as needed to palpate the muscles in a circumferential manner. Envisioning a clockface is useful as a frame of reference to localize examination findings, with the pubic bone at 12 o'clock and the anus and coccyx at 6 o'clock (Fig. 39.4). The levator ani can be palpated on both vaginal and rectal examinations from 1 to 5 o'clock on the left and 7 to 11 o'clock on the right, with the pubococcygeus located more anteriorly and the iliococcygeus located more posteriorly. Of note, the individual levator ani muscles are physically indistinguishable by palpation, and while they can be approximately identified based on general location of the examining digit, this detailed distinction often does not affect management. As with the superficial muscles, the deep PFMs are palpated for tone, tenderness, and trigger points. The presence of scar tissue or fibrosis should be noted as well as muscle defects, such as levator avulsion or sphincter gap.

The obturator internus is located just above 3 o'clock on the left and 9 o'clock on the right and can be distinguished from the levator ani by locating the arcus tendineus, which feels like a taut band similar to a guitar string (see Fig. 39.2), and palpating anteriorly to this structure. The obturator internus can more clearly be identified by having the patient activate the muscle, by rotating the hip externally against resistance. To do this, the examiner can place the free hand on the lateral aspect of the patient's knee and ask the patient to push the leg into the hand. This will cause the obturator internus muscle to bulge medially, which can be appreciated via internal vaginal palpation with the examining finger at the 3 or 9 o'clock position.

Rectal examination affords the ability to additionally assess anal sphincter tone as well as the levator ani, coccyx, and piriformis. The puborectalis, forming the innermost portion of the anal canal, can easily be appreciated on rectal exam, with the rest of the levator ani extending outward around it. The coccyx can be examined intrarectally to assess for tenderness, mobility, and anterior or lateral deviation. A Tinel sign can be elicited by tapping over the pudendal nerve as it courses inferior to the ischial spine and may provoke pelvic floor or perineal paresthesias.



• **Fig. 39.4** A clockface diagram is useful as a frame of reference for pelvic floor examination and is shown here superimposed on two views of the pelvic floor. (A) Pelvic floor from external point of view, as if the patient were oriented in the hook-lying position. (B) Reverse view, looking at the pelvic floor from the internal point of view. The pubic bone is at the 12 o'clock position, and the coccyx is at the 6 o'clock position. The left levator ani is located from 1 to 5 o'clock, and the right levator ani is located from 7 to 11 o'clock. The left obturator internus is located just above 3 o'clock, and the right obturator internus is just above 9 o'clock. (Redrawn from Prather H, Dugan S, Fitzgerald C, et al. Review of anatomy, evaluation, and treatment of musculoskeletal pelvic floor pain in women. *PM&R.* 2009;1:346-358.)

Internal Functional Examination of Pelvic Floor Muscles

Assessment of PFM function on internal vaginal and/or rectal examination is a key aspect of the pelvic floor examination. On internal exam, a voluntary contraction is felt as a tightening, lifting, and squeezing action under the examining finger and can be rated in terms of endurance, repeatability, and strength.[73] Strength can be assessed using the modified Oxford scale.[72] Similar to manual muscle testing for limb muscles, the scale ranges from 0 to 5, with 0/5 signifying "absent" contraction and 5/5 indicating that the patient is able to "lift, tighten, and maintain for 10 seconds" (Table 39.3). The patient can be cued to perform this motion by asking the patient to use only the PFMs to make the motion as to hold back urine or stool. When examining vaginally, strength testing should be performed bilaterally adjacent to the urethra, especially in patients with neurologic deficits such as hemiplegia. Relaxation postcontraction of the PFMs should occur upon asking the patient to release the muscles after performing a voluntary contraction and is felt as a termination of the contraction as the muscles return to their resting state. Endurance is tested by asking the patient to hold a full contraction for 10 seconds. Coordination is tested by performing "quick flicks" or asking the patient to contract and relax the PFMs rapidly. The examiner can have the patient cough to look for the presence or absence of involuntary contraction. The examiner can then ask the patient to replicate the motions of urination and defecation to look for presence or absence of reverse Kegel, which would be perceived as descent and outward movement of the PFMs.

Interpretation of Findings

The findings on examination can lead directly to a diagnosis of pelvic floor dysfunction, categorized earlier in Overview of Terminology.[73] Disorders of increased PFM tone are characterized by high tone and may feature potential impaired relaxation postcontraction, deficient or absent reverse Kegel, tender points in taut bands of muscle (areas of localized increased tone), twitch response, and reproduction of patient's usual pain on palpation. Of note, PFMs with increased tone can also exhibit impaired contractile strength and endurance caused by shortened muscle length, impairing their ability to function appropriately to allow for normal bowel and bladder mechanics.[69] In addition, these patients often have pain in the PFMs as a result of chronic hypertonicity.

Disorders of decreased PFM tone are characterized by low tone and may feature introital or anal gaping, levator or sphincter muscle defects, and absent or impaired PFM contractile function in terms of strength and/or endurance. In disorders of PFM pain, one notes normal PFM tone with tenderness of PFMs. In disorders of PFM coordination, dyssynergia appears on examination as paradoxic contraction or lack of relaxation of PFMs when relaxation is required, such as when cued to bear down and replicate

TABLE 39.3 Modified Oxford Scale Used to Grade Internal Manual Muscle Testing of the Pelvic Floor Muscles

Grading	Contraction	Lift or Tighten
0/5	No	No
1/5	Flicker	No
2/5	Weak	No
3/5	Moderate	Some lifting/tightening, contraction visible
4/5	Good	Holds for 5+ sec
5/5	Strong	Holds for 10+ sec

the voiding or defecation motion, and vaginismus and anismus are characterized by increased tone and lack of relaxation with attempted vaginal or anal penetration, respectively, often with associated pain.

Symptoms of Pelvic Floor Dysfunction

Symptoms of pelvic floor dysfunction include myofascial pelvic pain, urinary and/or fecal incontinence, urinary urgency and frequency, and functional constipation. Patients can present with one or several of these symptoms depending on the type and severity of their pelvic floor dysfunction. For example, patients with functional defecation disorders should be screened for urinary urgency and frequency because both can be a result of disorders of increased PFM tone or disorders of PFM coordination. Overall, these symptoms will improve with PFPT and other conservative treatment interventions. Of note, if patients have comorbid neurogenic bladder or bowel, PFPT may not be as effective in improving their urinary or bowel symptoms compared with patients without a neurogenic etiology. However, PFPT with a focus on certain techniques such as Credé or Valsalva voiding maneuvers (coordinating PFM relaxation with bear down or manual pressure on bladder) and neurogenic bowel and bladder education can still be attempted given that it is a relatively benign treatment intervention. Providers should engage in a conversation with these patients to set realistic expectations for PFPT duration and outcomes in the setting of underlying neurologic impairment.

Pelvic Floor Myofascial Pain and Dyspareunia

Definitions and Etiology

As noted, pelvic floor dysfunction, either with or without increased PFM tone, and/or with impaired PFM coordination, can result in pain. This general phenomenon, which can be called pelvic floor tension myalgia, pelvic floor myofascial pain syndrome, pelvic floor myalgia, or myofascial pelvic pain, will be referred to here as pelvic floor myofascial pain (PFMP).[28,73] In this section, we discuss pain of less than 6 months' duration (CPP is a distinct entity and will be covered later in this chapter). PFMP affects both sexes and can result from underlying weakness, overuse, trauma, injury (i.e., vaginal delivery, particularly with perineal laceration or obstetric anal sphincter injuries), compensatory injury (as is the case with PFMP in patients with lumbopelvic or hip pathology), or can be iatrogenic (resulting from pelvic surgery).[12,28] PFMP is defined as a regional condition of myofascial pain and tightness caused by chronic muscle contraction and is characterized by tender points, taut bands, and trigger points within the PFMs.[28,73] Given these features, PFMP is generally associated with increased pelvic floor tone. Myofascial trigger points classically cause pain referral and a twitch response with pressure. They develop from functional events such as overuse, repetitive strain or injury, and dysfunctional posturing. Trigger points can also be a sign of a viscerosomatic convergence, in which visceral pathology influences somatic function and can induce muscle guarding reflex.[100]

Epidemiology

PFMP is likely underdiagnosed and undertreated because of a lack of education on detailed pelvic floor examination and management of PFM pain in primary care (obstetrics-gynecology, family medicine, internal medicine) and physical medicine and rehabilitation residency training programs.[1,2] PFMP often coexists with other pelvic floor disorders and painful pelvic conditions, acting as a distinct etiology of pain. The prevalence of PFMP on transvaginal palpation among a sample of 912 patients with symptoms of a pelvic floor disorder was found to be 85% in a 2019 study, with 50% found to have severe PFMP.[124] Another study showed that more than 50% of females with endometriosis were noted to have PFM spasms compared with 17% of controls without endometriosis.[23] In another study, 13% of males with chronic prostatitis/CPP syndrome were found to have PFM tenderness compared with 7% of controls.[165]

Diagnosis and Physical Examination

The diagnosis of PFMP is made clinically by a combination of comprehensive history and pelvic floor physical examination. A thorough history of the pain complaint, including descriptors of pain onset, location, character, and severity, should be obtained. In addition, one should make note of pertinent medical comorbidities, medications, and prior procedures or surgical interventions. If applicable, a thorough gynecologic history (including age at menarche, regularity and characteristics of menstrual cycle, menopausal status, prior gynecologic procedures/surgeries) and obstetric history (including gravida/para status, history of labor prolongation, birth-related trauma including lacerations or vaginal instrumentation) should also be ascertained.[21,28] Patients should be assessed for comorbid chronic pain conditions (see Chronic Pelvic Pain, later), pelvic girdle pain (PGP), hip and low back pain, and chronic prostatitis in males and be screened for comorbid depression and anxiety.[21,28,102,108] A thorough review of any related body systems should also be obtained (including neurologic, musculoskeletal, urologic, gastrointestinal, dermatologic, infectious, oncologic, and psychiatric). Providers should be prepared to broach the subject of prior emotional, physical, or sexual trauma given the correlation between prior trauma and pelvic pain.[193] Elements of the history that flag PFMP include "deep" or "internal" pain, exacerbation with prolonged sitting, and associated symptoms, including dysuria, urinary frequency, urinary urgency, dysmenorrhea, dyspareunia, dyschezia, and painful defecation (particularly in the presence of negative obstetric-gynecologic, urologic, and gastroenterologic evaluations).[28,69] In obtaining a thorough history of sexual pain, providers should determine the location of pain, whether it occurs on initial penetration, deeper penetration, is position dependent, or occurs with orgasm or ejaculation. Patients must often be prompted with direct questioning to elicit such history because of the intimate nature of PFMP.

After obtaining a detailed history, a detailed pelvic floor physical examination is recommended, which should be done in conjunction with a thorough neurologic and musculoskeletal examination of the lumbosacral spine, pelvic girdle, hips, and lower extremities to provide a context for the etiology of the patient's symptoms. (This exam is outlined earlier in the chapter in detail.) When the chief complaint is PFMP, special attention should be paid to the palpatory exam, assessing for increased tone, tenderness, and trigger points in the PFMs as well as the surrounding musculature of the abdomen/suprapubic region, gluteal region, and lateral hips. Palpating the abdomen for diastasis recti, which can contribute to PFMP in postpartum females, is also important.[20,69] A physiatrist should maintain a low threshold for referral to a gynecologist, urologist, colorectal surgeon, or gastroenterologist for further evaluation if the exam raises suspicion for nonmusculoskeletal or neurologic etiologies of pain.

Diagnostic imaging can be useful in ruling out other musculoskeletal causes of lumbopelvic pain. Conventional pelvic and lumbar radiographs are useful to evaluate the structural integrity of the spine and pelvis. Based on physical examination findings, the physician may consider the addition of hip radiographs. Magnetic resonance imaging (MRI) and/or computed tomography (CT) of the spine, hip, or pelvis may be useful to rule out urgent causes of pelvic pain, including herniated lumbosacral disc, lumbopelvic fracture, infection, and malignancy. Ultrasound is an emerging imaging technique, with increasing applications to evaluate soft tissue, PFMs, joints (hip, pubic symphysis, etc.), organs, and neural structures.[6,58]

Treatment

Management of PFM pain should involve a multidisciplinary and individualized approach, given the broad range of overlapping diagnoses and multitude of pain generators across body systems. Depending on the patient, a multidisciplinary team may include physiatry, PT, obstetrics-gynecology, urogynecology, urology, gastroenterology, colorectal surgery, neurology, psychiatry, and psychotherapy or sex therapy. Given the strong correlation between anxiety and history of abuse or trauma and PFMP, particularly dyspareunia, it is crucial to ensure that patients' psychiatric comorbidities are being simultaneously managed via counseling and/or medications.[61,166] PFPT is the mainstay of rehabilitation treatment for PFMP and should be the first-line treatment offered.[112,177] The overall goals of PFPT are to resolve muscle imbalances and restore optimal PFM length-tension relationship, improve function, and reduce pain. PFPT is performed by a skilled physical therapist with specialized training and experience in treating pelvic floor dysfunction to provide the optimal assessment and treatment. Therapy should be individualized to the patient's needs and specific examination findings, and it should emphasize education, behavioral training, and active patient participation in learning a home exercise program in addition to passive treatment modalities. Because PFMP tends to be associated with increased PFM tone, pelvic floor downtraining techniques are often beneficial, including training on diaphragmatic breathing exercises, active PFM lengthening or reverse Kegels (with potential use of biofeedback via probe or surface electromyography [EMG][89]), myofascial release, and progressive relaxation techniques, along with use of vaginal and/or rectal dilators.[69,116,119,162] Dilators, in particular, play an essential role in allowing for gradual progressive lengthening of PFMs, desensitizing the muscle as well as soft tissue to stretch, and reducing pain and anxiety with penetration.[113] Other common elements of PFPT include stretching and strengthening exercises, muscle energy technique, strain-counterstrain, scar and connective tissue mobilization, visceral mobilization, desensitization, mirror therapy, and imagery. Physical therapists may also use modalities such as heat, ice, biofeedback, dry needling, rectal balloon catheter training, and electrical stimulation.[28,108] Because the PFMs are intimately related to the anatomic structures of the spine, pelvic girdle, hips, and core musculature, the pelvic therapist takes into account the biomechanics of these related areas and can prescribe an individualized approach to restore normal movement patterns, joint range of motion, and muscle strength.[143]

Pharmacologic treatment for PFMP resembles what is used for myofascial pain in other body regions, with the exception of some intravaginal or intrarectal medication options. We will focus in this section on treatments for musculoskeletal pain, including discussion of neuropathic pain agents and medications that reduce pain (treatment of concomitant mood disorders and central pain sensitization can be found later in Chronic Pelvic Pain). Acetaminophen is first line for analgesia given its favorable safety profile. Nonsteroidal antiinflammatory drugs (NSAIDs) are often used to treat acute pain but are limited from long-term use by gastrointestinal, renal, and cardiovascular side effects as well as the risk of bleeding and potential for interaction with other medications, such as aspirin. Care should be taken to avoid the use of opioid pain medications except as a last resort for patients with severe pain under the guidance of a pain physician when other treatment options have failed.[28] Orally administered muscle relaxants (e.g., cyclobenzaprine, baclofen, diazepam) may be helpful, especially for painful nighttime muscle spasms, but are limited by the side effect of sedation and are not recommended for long-term use.[28,70]

Lubricants and vaginal moisturizers can be helpful in dyspareunia. Use of topical or vaginal estrogen cream may also be helpful in addressing the role of a hypoestrogenic state in vulvovaginal pain. Topical anesthetic agents (e.g., lidocaine gel) may have a role in the treatment of PFM pain and facilitation of PFPT and dilator use. Other topical agents, such as topical tricyclics or NSAIDs, have a plausible mechanism for efficacy but are not well studied.[13,66]

Compounded vaginal or rectal suppositories containing muscle relaxants (such as diazepam or baclofen) are often prescribed by clinicians to relieve flares of pelvic floor pain or for use before pelvic floor PT, sexual intercourse, or at night before going to sleep. Recent randomized controlled trials (RCTs) for vaginal diazepam have not shown benefit in terms of vaginal muscle tone or pain.[53,90,172] Patients should be counseled that there is systemic uptake of vaginal diazepam, meaning that they may experience sedation as well as risk for dependency or addiction.[42,107]

When the previously mentioned rehabilitation and medication treatment interventions do not provide adequate relief from PFMP, interventions such as pelvic floor dry needling, myofascial trigger point injections, and pelvic floor botulinum toxin injections can be considered to reduce pain and increase participation in therapeutic exercises. Dry needling is a commonly used modality for treatment of myofascial trigger points throughout the body, and it was seen in a case series to reduce pain and associated disability when applied to the PFMs of patients with PFMP.[76] Trigger point injections to the PFMs can be performed transvaginally, paravaginally, or transperineally, and they can contain local anesthetic as well as steroid.[28] One retrospective study of 101 patients with pelvic floor pain and hypertonicity showed significant improvement in pain scores after transvaginal trigger point injections.[19] Pelvic floor botulinum toxin injections can be performed using similar approaches to trigger point injections, and they work by inducing temporary muscle weakness and inhibition of the release of pain-sensitizing peptides. In a 2022 meta-analysis of eight studies in which 289 participants received pelvic floor botulinum toxin A injections, there was a statistically significant improvement in visual analog scale pain scores.[168] It is important to note that the use of botulinum toxin for trigger point injections remains an off-label indication and may be of considerable financial cost to the patient. If the patient also complains of posterior pelvic myofascial pain, a trial of an ultrasound-guided piriformis or obturator internus muscle trigger point injection may be indicated. Use of injections should be approached judiciously because more research is still needed. Injectable interventions should not be used in isolation but as part of a comprehensive rehabilitation plan to aid in diagnosis and progress toward the goals in therapy—the reduction of pain and improvement of function.

Urinary Incontinence

Definitions and Etiology

UI, defined as the involuntary leakage of urine,[86] can be divided into three main types: stress urinary incontinence (SUI), urge or urgency urinary incontinence (UUI),[125] and mixed urinary incontinence (MUI). SUI occurs in situations with increased intraabdominal pressure, as during coughing, laughing, sneezing, or physical exertion. If there are deficiencies in the PFMs, urethra, bladder, and/or sphincter such that it is difficult to maintain urethral closure pressures during times of increased intraabdominal pressure, then urinary leaking occurs. The etiology is multifactorial and has been shown to be related to pregnancy, vaginal delivery, pelvic surgery, POP, neurologic causes, active lifestyle, obesity, and aging. UUI, also referred to as overactive bladder (OAB) with incontinence, is involuntary leakage accompanied by or immediately preceded by the sudden onset of an urge to void that cannot easily be deferred. The compilation of symptoms, including urinary urgency with or without incontinence, daytime frequency, and nocturia without obvious pathology such as urinary tract infection, is termed OAB.[98] The cause of UUI may be idiopathic, neurogenic, inflammatory, iatrogenic, or behavioral. UUI can occur when an involuntary detrusor contraction overcomes the sphincter mechanism or when poor bladder compliance results from loss of the viscoelastic features of the bladder. Neurologic processes can have a variety of functional changes on the bladder and bladder outlet system leading to incontinence from two main issues: failure to store and failure to empty. (See Chapter 21 for a more detailed understanding of neurogenic bladder.) Processes that change the bladder tissue, such as inflammation (recurrent urinary tract infections or interstitial cystitis) or radiation, can lead to UUI. Behavioral patterns, such as high-volume fluid intake, can also result in UUI. MUI occurs when patients experience both SUI and UUI symptoms.

Epidemiology

UI is by far more common in females. A study using data from the 2015–18 National Health and Nutrition Examination Survey found the overall prevalence of UI in adult US females over 20 years old to be 61.8%.[138] Within this group, 37.5% endorsed SUI, 22% UUI, 31.3% MUI, and 9.2% reported unspecified UI.[138] These findings are similar to prior studies that found stress incontinence to be more prevalent than urge or mixed incontinence, with all types increasing with age.[92,94,127] Reported risk factors for UI include obesity, lower education level, lower physical activity level, diabetes, high levels of anxiety or depression, decreasing independent functional status, positive smoking history, prior hysterectomy, and prior vaginal birth.[138] One study, which looked at vaginal delivery and its effect on the prevalence of different types of UI in females, found a 15.3% rate of SUI, a 6.1% rate of UUI, and a 14.4% rate of MUI.[96] A 2020 meta-analysis looked at high-impact sport female athletes aged 18 to 45 years and found the prevalence of UI to be 25.9%.[140]

Diagnosis and Physical Examination

Physiatrists often discover UI as part of screening questions for back pain or on review of systems. This is an essential line of questioning because fewer than 50% of those suffering from this condition will seek treatment. If a patient admits to this condition, it is important to review the medical history, noting any neurologic disease. The physician should determine whether the patient has SUI, UUI, or MUI, with questions regarding the timing of accidents and their relation to activity and urgency symptoms. If a patient with UI has presented to the office because of back pain, it is necessary to rule out surgical emergencies, such as cauda equina syndrome. Patients who do not have neurologic deficits should be further examined by a physiatrist who is comfortable performing a pelvic floor examination or referred to a urologist for additional workup.

A thorough neurologic, lower extremity, and pelvic physical examination, as described previously, can help distinguish between neurogenic causes and PFM contributions to incontinence. For UI in particular, assessment for gross POP is needed. It is important to note POP in a patient with back pain and incontinence because this may be a source of vague, achy low back pain. The PFMs are often found to have decreased tone and weakened strength in SUI and increased tone in UUI. However, increased PFM tone can also be seen in SUI/MUI.[162] PFMs with increased tone can lead to dysfunctional and weakened coordinated contractions that result in urinary leaking in settings of increased abdominal pressure. The integrity of the vaginal mucosa should also be noted because menopausal hypoestrogenic state has been associated with UI.[109] Determination of when to refer for advanced testing—such as cystoscopy, postvoid residuals, and urodynamics—should be made on an individual basis.

Treatment

Management for UI includes lifestyle modifications, PFPT, pharmacologic, and procedural/surgical treatments if necessary. Conservative treatment for UI often results in improvement of symptoms. Treatment options vary for SUI, UUI, and MUI, but all respond well to behavioral modifications and rehabilitation interventions.

Behavioral and lifestyle alterations include changes in diet to reduce consumption of bladder irritants, smoking cessation, regulating fluid intake, and bladder training. In a 2018 update to the Agency for Healthcare and Research Quality's systematic review on treatment for UI, there was moderate- to high-strength evidence that first-line intervention behavioral therapy generally resulted in better UI outcomes (cure, improvement, satisfaction) than second-line interventions (medications).[16] Reduction of more than 5% body weight has led to a 47% decrease in incontinence episodes vs. 28% with education alone.[174] Bladder training usually consists of two parts, timed voiding and urge suppression techniques; it can take approximately 12 weeks to see improvement. The overall goals of bladder training are to prolong the time between voids and to void before experiencing a sense of urgency or UUI. Using a bladder diary to identify the shortest interval between voids, patients are instructed to increase this time by approximately 15 to 30 minutes until there is a period of approximately 3 hours between voids. Urge suppression can temporarily reduce the intensity of a bladder contraction by causing reflex inhibition of the parasympathetic nerves acting on the detrusor muscle. Upon feeling an urge, the patient should stop and/or sit, perform five to six quick voluntary contractions of the PFMs, take a deep breath, relax until the urge passes, and then walk to the restroom normally.

As discussed earlier, patients with SUI caused by decreased PFM tone benefit from Kegel exercises (i.e., exercises focused on voluntary contraction of PFMs). Studies have shown PFM training with or without biofeedback or electrical stimulation reduces UI.[27] There is no significant difference when PFM training is compared with biofeedback; however, both interventions are superior to controls.[9] A strengthening program for the PFMs increases support for the bladder and urethra and helps maintain urethral closure pressures. One study found reduced bladder neck mobility during coughing and hypertrophy of the urethral sphincter after a 12-week PFM training course.[122] These changes are thought to contribute to reduction in SUI frequency.[122] As strength improves, patients can be taught to make a PFM contraction an automatic response when they are anticipating an increase in intraabdominal pressure (as with a sneeze) and to perform such contractions during functional activities. The long-term effect of PFM strengthening for SUI symptom improvement ranges from 41% to 85%.[27]

Of note, if patients with increased PFM tone are found to have SUI, they should first undergo a downtraining program to establish normal baseline PFM tone before learning how to perform PFM contraction in settings of increased intraabdominal pressure to prevent incontinent episodes.[162] Furthermore, we recommend to avoid use of electrical stimulation in patients with increased PFM tone because it would worsen their tone and be counterproductive to their treatment plan.

Second-line treatment for UI is considered to be medications. There are currently no approved drugs for SUI in the United States. The mainstay of pharmacologic management of OAB and UUI are anticholinergic medications that decrease urgency and detrusor instability by blocking parasympathetic nerves of the bladder. However, their use and efficacy are often limited by side effects, such as constipation, drowsiness, and dry mouth. Topical vaginal estrogen has been shown to improve continence, especially in females with incontinence, urgency, and frequency related to genitourinary syndrome of menopause or other hormone-depleted states.[50,144,189] Examples of decreased estrogen states in premenopausal females include but are not limited to postpartum deficiency, use of oral contraceptives or sex-affirming hormone therapy, ovarian failure after radiation, and surgical removal of bilateral ovaries.[50]

When conservative measures have failed, other treatment options can be considered with specialists such as urogynecologists and urologists. Procedures for UUI that may be considered are intravesical botulinum toxin and neuromodulation to decrease detrusor muscle contractility. Neuromodulation could also target stimulation of the tibial nerve/S3 sacral nerve root. Advanced treatment options for SUI may include inserting periurethral bulking agents to increase compression on the urethral lumen or surgical placement of urethral slings to provide improved urethral support.

Urinary Urgency and Frequency

Definitions and Etiology

Urinary urgency is defined by the ICS as "the complaint of a sudden compelling desire to pass urine which is difficult to defer."[86] Frequency of urination more than every 2 to 3 hours can be considered abnormal. The symptoms of urgency and frequency are similar to those of UUI (see earlier). There are also neurogenic, inflammatory, hormonal, age-related, and myogenic causes of urgency/frequency symptoms. The pathophysiology of neurogenic urgency and frequency is usually attributable to detrusor overactivity as a result of disruption of the complex micturition reflex at the cerebral, pons, or corticospinal level (see Chapter 21).

Inflammatory causes of urinary urgency and frequency include urinary tract infections, in which increased inflammation in the bladder leads to sensory afferent upregulation, ultimately causing detrusor muscle overactivity. Similarly, estrogen deficiency leads to vaginal and urethral atrophy of the superficial and intermediate layers of the urethral mucosal epithelium, which in turn causes urethritis, diminished mucosal seal of the urethra, decreased bladder compliance, and irritation, which may bring on increased infections and/or UUI. The normal aging process also leads to increased bladder contractility and decreased bladder compliance resulting in urgency and frequency, respectively.

Patients with urinary urgency and frequency in the absence of bladder or neurologic pathology may have these symptoms as a result of increased PFM tone (see earlier). Those in this category will have dysfunctional voiding with a sense of urgency and/or frequency with a constant need or an exaggerated urge to void. Patients with increased PFM tone can have difficulty with relaxation postcontraction or can have involuntary PFM contractions during voiding (dyssynergia). Other symptoms of pelvic floor dysfunction resulting from increased PFM tone include postvoiding pain, urethral pain, hesitancy, or incomplete bladder emptying.

Epidemiology

As stated previously, the term OAB refers to symptoms of urinary urgency/frequency with or without the presence of incontinence. This must be taken into consideration when reviewing epidemiology findings. The NOBLE study in 2002 found a US prevalence of OAB without UUI to be 10.4%.[171] Males were more likely to have OAB without UUI (13.4%) than with UUI (2.6%).[171] The EpiLUTS study reports various combinations of OAB symptom prevalence among both sexes in the United States (15.6% of males reported urgency alone, and 7.2% reported urgency and frequency[52]; 10.9% of females reported urgency alone, and 4.1% reported urgency and frequency[52]). Other terminology has also been emerging, such as lower urinary tract symptoms (LUTS), which includes UI and OAB, that makes more recent prevalence data on urinary urgency/frequency without incontinence more difficult.[51]

Diagnosis and Physical Examination

Patients presenting with urinary urgency and frequency symptoms would benefit from a multidisciplinary evaluation, including urology or urogynecology, for a complete workup of their voiding dysfunction, including postvoid residuals and urodynamic testing. This can help guide the differential and help put findings of the pelvic exam by physiatry into context, performed as described previously. Males should have a thorough prostate examination with urology. As with UI, a thorough evaluation of patients with urinary frequency and urgency includes a full medical history and physical examination to assess for pathologic causes (see earlier). Diagnosis of increased PFM tone can be made during a musculoskeletal pelvic floor examination, with findings described in the physical exam section.

Treatment

First-line treatment is the same as for UUI—behavioral therapy and PFPT, often done in concert because they are considered noninvasive and reversible and have good literature support[9,16] (see earlier). Other treatments for urinary urgency and frequency are the same as those discussed earlier for UUI and include lifestyle modifications and consideration of medications and procedures.

Unfortunately, the literature is limited with regard to PFPT treating urinary urgency/frequency symptoms independently, without incontinence or pelvic pain, because these symptoms often present together in the clinical setting. In addition, many studies focus on PFM uptraining in the setting of decreased PFM tone and do not comment on PFPT treatment of urgency/frequency in patients with increased PFM tone.[9] The goal of PFPT in patients with urgency and frequency, especially when they have increased PFM tone, is to teach patients how to relax the PFMs during voiding. This relaxation process is referred to as downtraining. PFM training focuses first on developing an awareness of muscle contraction vs. relaxation with biofeedback. The focus should then turn to teaching volitional relaxation of the PFMs in the context of voiding. Perineal, rectal, or vaginal biofeedback can be especially useful for bringing muscle tension to a conscious level. In addition, to facilitate relaxation, patients can be taught to relax the muscle for a longer period of time compared with a contraction, usually in a 1:2 ratio.

Fecal Incontinence

Definitions and Etiology

FI is an often distressing condition defined as the recurrent involuntary loss or uncontrolled passage of solid or liquid feces.[57,86] The broader term *anal incontinence* encompasses FI in addition to the involuntary loss of mucus or flatus. FI can be further categorized as (1) passive incontinence (fecal soiling without sensation or warning, or difficulty wiping clean) and (2) urge FI (loss of stool associated with sudden urge).[57,86] FI occurs when the PFMs/anal sphincter cannot hold back the passage of stool. The etiology can be gastroenterologic, neurogenic, muscular, or iatrogenic. It can occur as a result of rapid gastrointestinal transit, severe constipation with overflow diarrhea, neurologic injury, anal sphincter injury, or weakness or incoordination of PFMs (which can occur in disorders of high pelvic tone, low pelvic tone, or impaired coordination of PFMs). Patients with gastrointestinal disorders, such as irritable bowel syndrome (IBS) or inflammatory bowel disease, have a higher likelihood of having FI.

Epidemiology

The prevalence of anal incontinence is reported between 7% and 15% of the general population, notably higher among older adults and females.[57] FI is also prevalent in high-level athletes, although to a lesser extent than SUI. One study of 393 female athletes found a prevalence of anal incontinence of 14.9% and reported that 8% of all high-level athletes had to wear a pad for protection as a result of significant FI.[185] FI is more common in patients who have weakness or injury to the anal sphincter. Obstetric factors are likely to play a large role: Sphincter laceration, use of forceps, midline episiotomy, and intrapartum pudendal nerve damage have all been shown to contribute to the development of subsequent FI.[12] Pelvic surgeries can also predispose to the development of this disorder. FI has been reported as a complication in 33% of hemorrhoidectomies, 11% of sphincterotomies, and 9% to 32% of radical prostatectomies.[134] Pelvic radiation can lead to the development of FI in 14% to 46% of patients.[134]

Diagnosis and Physical Examination

As with UI, it is important for the physician to inquire about the presence of FI because patients are often reluctant to discuss the subject; it is estimated that less than 30% of the people who struggle with FI discuss the problem with their health care providers.[134] It is important to determine whether there is loss of liquid or solid stool, mucus, or flatus. The timing and frequency of episodes, the volume of stool loss, and the ability of the patient to sense the involuntary passage of anal contents are all necessary elements of the history. The physical examination should include a thorough neurologic and spine examination (see earlier) as well as a digital rectal examination (DRE) to assess for sphincter tone and sphincter defects. PFM tone, strength, and endurance are also important and can be assessed on rectal exam. Although physical examination is the mainstay of diagnosis, additional diagnostic tools can be used to further evaluate FI. Anorectal manometry (ARM) measures the strength and endurance of the anal sphincter at rest and with contraction, and assesses for dyssynergia, rectal sensation, reflexes, and compliance. Of note, DRE is quite accurate relative to ARM for assessing anal resting tone and squeeze function and for identifying dyssynergia.[145] Pudendal nerve terminal motor latency (PNTML) testing can be performed as part of electrodiagnostic testing,[153] but this is rarely used clinically. Endoanal ultrasound is an effective tool for evaluating the integrity of the internal and external anal sphincters.[154] From a pelvic rehabilitation standpoint, physical examination is sufficient to identify abnormalities of muscle tone and coordination that could benefit from pelvic PT.

Treatment

Treatments include conservative measures such as dietary modification, medication, and pelvic floor rehabilitation as well as more invasive approaches such as the use of perianal injectable bulking agents, sacral nerve stimulation, and surgery.[33] A stepwise approach to treatment should be taken to minimize injury. Pelvic floor rehabilitation has been used successfully in the treatment of FI and can produce significant functional and quality of life benefits for patients.[191] Conservative treatment, including education, medications, behavioral strategies, and pelvic floor exercise, has been shown to restore fecal continence in up to 25% of patients, and biofeedback therapy has been shown to produce treatment satisfaction in 76% and fecal continence in 55% of patients.[191]

Only a few studies have looked at which types of patients will most likely benefit from a rehabilitation approach to treatment for FI. Good sphincter function and mild to moderate symptomatology are considered more favorable prognostic factors.[31] Disruption of the anal sphincter, spinal cord injury or other neurogenic disorders, severe impairment of rectal sensory function, cognitive impairments, severe depression or other mental illness, and age less than 6 years are all thought to be predictors of poor response to biofeedback and other rehabilitation treatments.[46]

Pelvic floor rehabilitation techniques for the treatment of FI include bowel management education, PFM training, biofeedback therapy, the use of electrical stimulation, manual myofascial release, and connective tissue mobilization techniques. The different rehabilitation techniques can be used independently but more frequently are used in conjunction with one another in a multimodal approach to produce the maximum benefit for the patient.[158]

The incorporation of lifestyle education into the therapeutic treatment program is of vital importance for patients with FI. Education is an important part of the PFPT process, but it can also be conducted within the physician's office before the initiation of pelvic PT. It is important to instruct patients about optimal fluid intake and dietary adjustments. Fiber supplementation—particularly soluble fiber, which reduces the liquidity of stools by forming a gel—has been shown in trials to improve FI symptoms over placebo.[48] The efficacy of soluble fiber in FI may also be comparable to antidiarrheal medications, such as loperamide, which can be of benefit but have the potential to cause constipation.[48] Behavior modification can also be explored with patients, including training on the establishment of a predictable pattern of bowel evacuation, timing of defecation relative to activities to limit incontinent episodes, techniques to reduce straining, proper defecation posture when sitting on the toilet, and fecal urge suppression techniques.[132,134,142]

The primary goal of PFPT for FI is to improve the efficiency of pelvic floor and anal sphincter muscles, which requires normalization of tone and improvement in range of motion, coordination, and endurance to effect a positive change in function with a decrease in symptoms. Additional goals include development of proprioceptive awareness and rectal sensitivity training (to either increase dulled urge sensation or quiet excessive urge sensation) and reduction of scar restriction to allow for improved muscle function.

There are three main approaches for the use of biofeedback as part of pelvic floor rehabilitation for FI.[133] Biofeedback therapy is most commonly used to improve the strength and endurance of the PFMs and/or the external anal sphincter. It has been theorized that this training is effective because it enables patients to hold the stool within the rectal vault for a longer time, allowing them to make it to a restroom with fewer accidents. The second treatment modality is to use biofeedback to improve rectal sensitivity or compliance. This type of treatment is typically done with sequentially inflated rectal balloons.[46] The rationale behind sensory retraining is to allow patients to detect smaller volumes of stool at an earlier time, again making it possible for them to reach a restroom before an accident occurs. The third biofeedback approach deals with coordination training for the anal sphincter.

When conservative treatment options have failed, procedures or surgeries can be discussed with specialists, such as colorectal surgery. Options that may be considered include perianal injectable bulking agents, sacral nerve stimulation, sphincteroplasty, muscle transposition procedures, artificial sphincter placement, and colostomy.[33]

Functional Constipation

Definitions and Etiology

Constipation can be divided into two major classifications, primary (or functional) and secondary.[37,97] Most relevant to the discussion in this chapter is functional constipation, which can be divided into subtypes of normal-transit constipation, slow-transit constipation (also known as colonic inertia), and rectal evacuation disorders (also known as outlet constipation). Outlet constipation is identified in the Rome IV criteria for disorders of gut-brain interaction as dyssynergic defecation, a type of functional defecation disorder.[60] It can occur as a result of PFM dysfunction, in which there is impaired relaxation or paradoxic contraction of PFMs during defecation (see discussion of dyssynergia, earlier). Most often, this incoordination results from the failure of the PFMs to relax appropriately during evacuation efforts. Other causes of outlet dysfunction constipation may include rectocele, enterocele, peritoneocele, and intrarectal intussusception. Common features of outlet constipation include prolonged or excessive straining, stools that are difficult to pass, and rectal discomfort.[97] Of note, slow-transit constipation and outlet constipation often coexist.

Epidemiology

Chronic constipation has been reported to affect approximately 10% to 15% of the general population, including all ages and both sexes.[15,97] Constipation is more common in females, with a female-male ratio of 2.2:1.[173] Constipation significantly decreases the quality of life and functional abilities of those affected and contributes to high health care costs to society.[97]

Diagnosis and Physical Examination

When taking the history of a patient with constipation, asking detailed questions about the patient's bowel habits—particularly defecatory frequency, stool consistency, difficulty evacuating—can help to differentiate between types of bowel dysfunction. A majority of patients with dyssynergic defecation will report excessive straining, incomplete evacuation sensation, hard stools, reduced stool frequency, and the need to perform digital maneuvers to facilitate evacuation (i.e., vaginal or perineal splinting or manual disimpaction).[145] Small, hard, pellet-shaped stools are indicative of poor colonic transit or excessive time in the rectal vault. Pencil-thin stools are a common finding with PFMs, with increased tone or disorders of coordination, such as dyssynergia or anismus.

Subsequently, a comprehensive evaluation, including rectal pelvic floor examination, should be performed (see Physical Examination section). DRE is the essential diagnostic tool in identifying dyssynergic defecation, but it is underutilized in practice.[145] As alluded to earlier, DRE is has been shown to have high sensitivity (75% to 93%) and specificity (59% to 87%) in detecting dyssynergic defecation.[37]

Diagnostic testing for functional constipation includes ARM (described earlier), balloon expulsion test, defecography, and colonic transit studies.[106] Of note, ARM lacks specificity in identifying dyssynergic constipation, as positive dyssynergic findings on ARM have been seen in nearly 90% of asymptomatic patients.[145] In the balloon expulsion test, often done as a part of ARM, a 4-cm-long balloon filled with 50 mL of water is placed in the rectum, the patient is instructed to expel the balloon, and this process is timed.[151] Defecography is another important tool in the diagnosis of constipation because it can distinguish between the many causes of outlet dysfunction constipation, including PFM dyssynergia, rectocele, and intrarectal intussusception. Defecography is a dynamic fluoroscopy study of evacuation performed in the sitting position after barium paste has been placed in the subject's rectum.[145] Magnetic resonance (MR) defecography is emerging as the modality of choice because of its superior image quality. Defecography tests are often used as an adjunct test when ARM and balloon expulsion tests are equivocal or nondiagnostic. Colonic transit times can be measured by scintigraphy or with abdominal radiographs after a patient has ingested radiopaque markers (i.e., sitz marker study) or a wireless motility capsule. Again, we feel that from a pelvic rehabilitation standpoint, physical examination is sufficient to identify abnormalities of muscle tone and coordination that could benefit from pelvic PT.

Treatment

The treatment of constipation depends greatly on the cause. Slow-transit constipation can benefit from increasing fluid intake, fiber supplementation, magnesium supplementation, and stool softener or laxative use. Over-the-counter osmotic laxatives, such as polyethylene glycol, or stimulant laxatives, such as bisacodyl and senna, can be utilized.[176] Rectal agents such as laxative enemas and intrarectal suppositories can also be of benefit. Prescription prokinetic medications, such as prucalopride, and prosecretory agents, such as linaclotide and lubiprostone, are often prescribed by gastroenterologists in refractory cases.[176] Before initiation of a new maintenance bowel regimen, we recommend having the patient perform a bowel cleanse. This will allow for improved effectiveness of the maintenance regimen and prevent overflow diarrhea or incontinence from retained hard stool in the colon. If such measures are not effective for severe slow-transit constipation, one could consider transanal irrigation, which allows for regular and predictable emptying of the rectosigmoid colon,[65] and ultimately surgical options such as subtotal or total colectomy with ileoanal anastomosis or colostomy.

Dysfunctional and dyssynergic defecation are ideally treated with PFPT. Treatment should include biofeedback and pelvic floor downtraining techniques (see earlier). The goals of therapy include education about dysfunctional defecation, coordination of increased intraabdominal pressures with pelvic floor relaxation during evacuation, therapist-assisted practiced defecation simulation with an intrarectal balloon, and use of anal dilators. The therapist will typically ensure that the patient is using proper defecatory posture while sitting on the toilet and often will instruct the patient to prop up the feet on a stool to elevate the knees above the hips, and lean forward slightly during defecation to allow for maximum relaxation of the PFMs.[128] A patient with a symptomatic rectocele can be taught to splint intravaginally with the thumb pressing backward toward the rectum to facilitate stool expulsion. There are several devices that can facilitate this splinting during defecation, including pessaries, vaginal splints, and external support devices that are either handheld or anchored to the toilet.[178] Patients with a significant enterocele, peritoneocele, or intrarectal intussusception often need surgical intervention such as a resection rectopexy. Options are limited for patients with severe dyssynergia or anismus who do not respond to PFPT. Botulinum toxin to the puborectalis (typically done under endoanal ultrasound guidance) can be considered, but the risk of development of subsequent FI must be taken into careful consideration.[106] The only surgical option for severe functional constipation refractory to the previously mentioned treatments is colostomy.

Neurogenic Causes of Pelvic Pain

Definitions and Etiology

Neural injury can be a source of pelvic pain and can coexist with pelvic floor dysfunction and PFMP. The iliohypogastric, ilioinguinal, genitofemoral, and pudendal nerves most commonly contribute to pelvic pain symptoms (see Fig. 39.3). Border neuralgia or border nerve syndrome is sometimes used to refer to neuralgia caused by an isolated or a combined dysfunction of the iliohypogastric, ilioinguinal, and/or genitofemoral nerves.[64] Pudendal neuropathy has also been implicated as a potential cause of UI and FI as well as sexual dysfunction. The inferior cluneal nerves are pure sensory nerves that branch from the posterior femoral cutaneous nerve. Inferior cluneal neuropathy is a rarer cause of neuropathic pelvic pain in the lower buttocks compared with pudendal neuralgia. The anatomy of these nerves has been reviewed earlier.

The iliohypogastric and ilioinguinal nerves are particularly susceptible to injury if a Pfannenstiel or low transverse incision is dissected too far laterally beyond the edge of the rectus abdominis muscles.[32,115] Neuroma formation is common after such nerve damage and can be a source of chronic pain.[114] The genitofemoral nerve can be damaged during surgery, including lymph node biopsy or vasectomy, and via compression during gynecologic surgery, often by poor placement of self-retaining retractors.[32,43] All three of these nerves are thought to be common causes of chronic groin pain after surgery for inguinal hernia repair.[81] Injury to the pudendal nerves during vaginal delivery has been well reported in the literature, and pudendal neuropathy has been implicated as a possible contributing factor to new-onset postpartum UI and FI.[49,74,161] The pudendal nerve can also be injured in pelvic surgeries (particularly with the use of vaginal mesh), in cases of pelvic trauma, with radiation therapy, bicycle riding, repetitive squatting/hip flexion, chronic straining to defecate, and during anal intercourse.[87,110]

Epidemiology

The overall incidence of ilioinguinal and/or iliohypogastric nerve injury after a Pfannenstiel incision has been estimated at 2% to 4%.[114,115,117] It has also been reported that up to 8% of patients may develop chronic inguinal pain, thought to be caused by inguinal nerve entrapment, after Pfannenstiel incision.[183] The incidence of chronic groin pain after hernia repair attributable to injury is thought to be as high as 16%, with 6% to 8% of all patients after herniorrhaphy having moderate to severe disabling symptoms.[44,149] Reported incidence of pudendal neuralgia in the general population has ranged from 1% to 4%[87]; however, it is thought the true incidence is significantly higher than reported in the existing literature because it is often unrecognized.[87]

Diagnosis and Physical Examination

The key to diagnosing pelvic neuropathic pain is first having a high index of suspicion. The history is often more helpful than the physical examination because overlap in the cutaneous distribution of these nerves often leads to minimal positive exam findings. The symptoms of iliohypogastric and ilioinguinal neuralgia typically include pain in the lower abdomen, groin, or pubis. There can be accompanying numbness, particularly if both nerves are affected simultaneously. Genitofemoral neuralgia is also often described as pubic or groin pain but can also be appreciated as labial or scrotal/testicular pain. If the femoral branch of the genitofemoral nerve is involved, there can be pain in the upper anterior thigh. Symptoms of pudendal neuralgia include perineal, clitoral/penile, labial/scrotal, and anal pain as well as pain medial to the ischial tuberosity. Symptoms are most often unilateral. The pain is most often described as burning, tingling, aching, stabbing, and electric shocklike, with less commonly reported numbness or paresthesia.[111] Pain is worse with sitting, dyspareunia is common, and patients may report a sensation of a foreign body in the rectum or perineum. Many patients with pelvic neuropathic pain will also have pelvic floor myofascial dysfunction; it can be difficult to determine on physical examination whether the muscles or the nerves are the primary source of the pain. This is particularly true with pudendal neuralgia because patients with significant PFM overactivity can develop resulting compression of the small, infiltrating branches of the pudendal tree, producing the characteristic pudendal burning sensation even in the absence of a true pudendal nerve lesion or entrapment.[87] It should be noted that, despite the anatomic function of the pudendal nerve, in many cases of pudendal neuralgia there is no objective loss of sensation or anal sphincter tone, and other neurologic examination testing such as the anal wink may remain normal in all but the most severe cases.[87] The Nantes criteria are a validated set of clinical conditions that have been proposed to assist in the diagnosis of pudendal neuralgia.[104] It should be noted that the Nantes criteria are thought to be sensitive but not specific, as patients with overactive pelvic floor dysfunction will also meet all the clinical criteria.[64] Clunealgia should also be considered in the differential when evaluating for possible pudendal neuropathy.

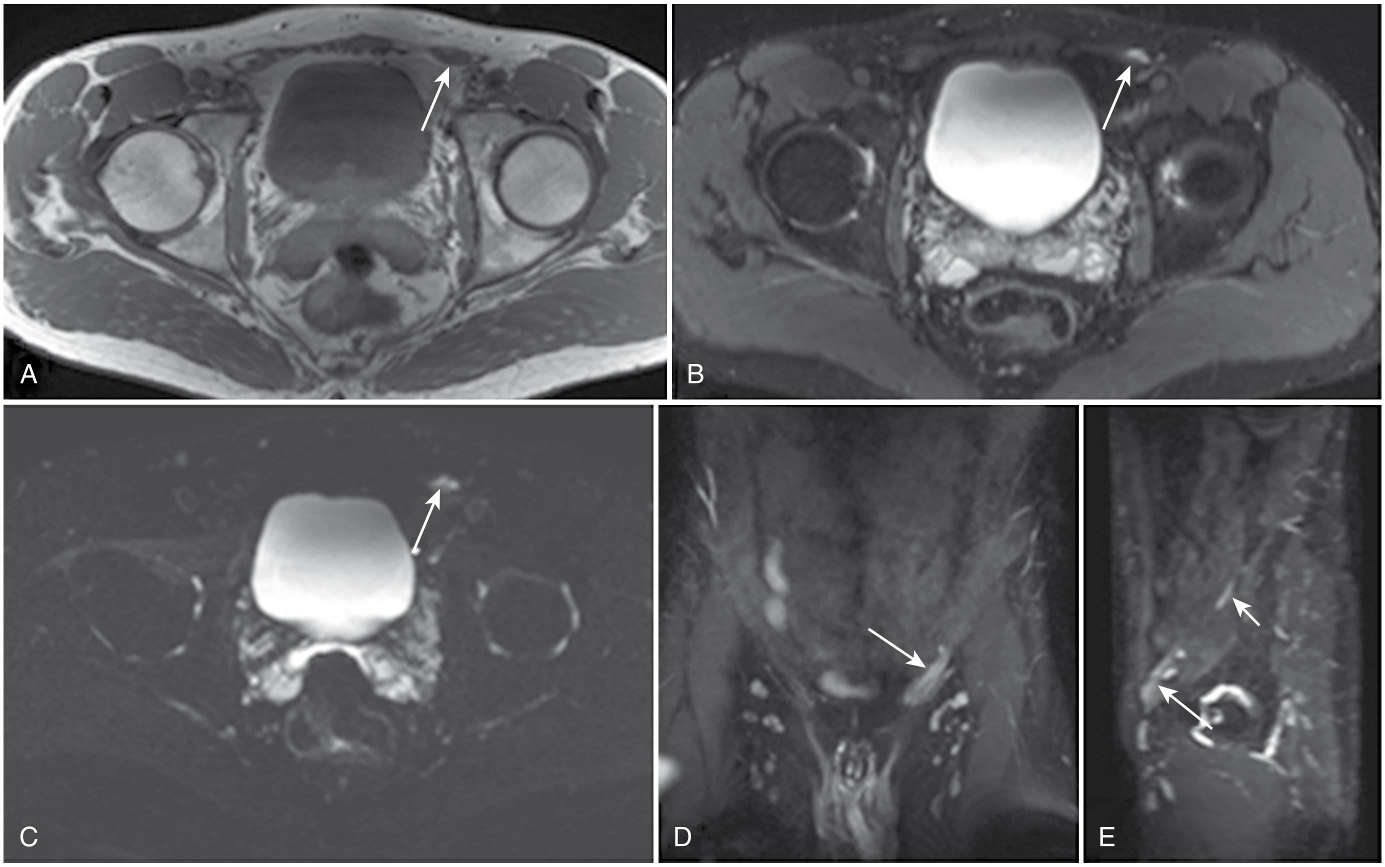
Given the similar territories the inferior cluneal nerve shares with the pudendal nerve, misdiagnosis is common, although again it is rarer compared with pudendal neuralgia.[64]

The iliohypogastric, ilioinguinal, genitofemoral, and pudendal nerves can be evaluated electrophysiologically by a number of different methods.[103] Unfortunately, these nerves are technically difficult to study; therefore these studies are not routinely used in clinical practice. PNTML testing can be obtained through the use of a St. Mark electrode, with nerve stimulation at the ischial spine and recording of muscle contraction response at the external anal sphincter.[49,161] The usefulness of PNTML has been questioned because it has been shown to have a high rate of inter- and intraobserver variability.[87] Needle EMG of the external anal sphincter or bulbospongiosus muscles can be performed.[8] More recent studies have utilized quantitative EMG of the external anal sphincter, which is thought to be easier to perform and have less operator bias.[79,80,190] Electrodiagnostic testing for pudendal neuropathy may be less well tolerated than standard nerve conduction studies and needle EMG of the extremities.[110] The bulbocavernosus reflex latency can also be obtained by stimulating at the clitoris or penis.[95,175] Somatosensory evoked potential testing has also been used to identify pudendal neuropathy.[110]

Diagnostic nerve blocks are generally considered the best way to make a diagnosis of neuropathic pelvic pain. A positive response to infiltration of a local anesthetic around the iliohypogastric, ilioinguinal, and/or genitofemoral nerves is thought to be a reliable indicator of etiologic correlation.[43,99] It is less clear that a positive response to a pudendal block is definitively correlated with true pathology because this nerve also innervates the sphincters and PFMs, and it has been proposed that patients with primary pelvic floor dysfunction will also respond favorably to pudendal nerve blocks.[104] It is always preferable to use imaging guidance for better accuracy when these diagnostic injections are being performed.[87] Diagnostic ultrasound in general is not particularly useful for evaluating nerve injuries about the hip and pelvis because these nerves are typically too small and deep to allow for long-segment exploration and good visualization.[120] MR neurography technology is rapidly becoming recognized as one of the most effective diagnostic tools for nerve injury and is thought to be far superior for nerve visualization than standard MRI.[45,167] MR neurography of the lumbosacral plexus is often able to demonstrate injury to the iliohypogastric, ilioinguinal, genitofemoral, and pudendal nerves, and can be successfully used to guide injections with more accuracy compared with ultrasound, fluoroscopic, and CT-guided perineural injections.[54] Fig. 39.5 shows MR neurography images of a left genitofemoral neuropathy.

Treatment

The first line of treatment for pelvic neuropathic pain may include PFPT (particularly for pudendal neuralgia) or medications. Neuropathic pain medications such as gabapentin, pregabalin, serotonin norepinephrine reuptake inhibitors (SNRIs), or tricyclic antidepressants (TCAs) may be of benefit. Compounded neuropathic pain creams and vaginal/rectal muscle relaxer suppositories are being prescribed more frequently in recent years,



• **Fig. 39.5** (A–E) Left genitofemoral neuropathy as seen on magnetic resonance neurography *(arrows)*. (Courtesy Dr. Avneesh Chhabra.)

although additional high-quality studies are needed to clarify concentration and efficacy.[160] Therapeutic injections of corticosteroid mixed with local anesthetic, delivered either as a single intervention or as an injection series, have been reported to be helpful for ilioinguinal, iliohypogastric, genitofemoral, and pudendal neuropathies.[87,170,179] As with diagnostic injections, therapeutic injections should ideally be performed under imaging guidance. Of note, a recent expert consensus states for pudendal neuropathy specifically, there is still a lack of literature demonstrating long-term analgesic effects of corticosteroid injection.[111] They also recommend not to repeat injections unless the first injection had a lasting improvement for several weeks.[111] Radiofrequency ablation and pulsed radiofrequency treatments for some of these nerves have also been described; however, they are not considered first-line treatment given the lack of studies and potential morbidity.[39,111,152,157] There have been a few descriptions of successful treatment of ilioinguinal or pudendal neuropathic pain via neuromodulation either at the level of the spinal cord or sacral plexus or of the individual nerves themselves, but at this time neuromodulation has not been studied extensively enough to be recommended for use in this patient population.[41,111,146,155]

Surgery can be an effective solution in refractory cases that have failed first-line treatment, particularly for chronic iliohypogastric, ilioinguinal, and genitofemoral neuralgia.[44,115,170] Neurectomy, also known as nerve resection or transection, is the preferred surgical option for these three nerves. Increasingly, a triple neurectomy of all three nerves is performed at the same time for patients with severe groin pain.[44] Rates of complete or moderate pain relief after neurectomy of the iliohypogastric, ilioinguinal, and genitofemoral nerves have been reported in the range of 66% to 100% of patients.[44,115,194] A recently published expert consensus on pudendal neuropathy states that pudendal nerve release is an effective treatment in 60% to 80% of cases with various surgical approaches.[111] Appropriate candidates should have the five Nantes criteria and failed first-line treatment.[111]

Pelvic Pain in Pregnancy and Postpartum

Definitions and Etiology

Many musculoskeletal changes occur during pregnancy to accommodate for the growing fetus and prepare the body for childbirth. As the fetus grows, there is an increase in maternal body mass, a lengthening of the abdominal muscles, an increase in lumbar lordosis and anterior pelvic tilt, which shifts the center of gravity anteriorly, and an increase in pelvic width.[163] Hormonal changes—including progressive elevations in estrogen, progesterone, and cortisol through each trimester as well as elevation in relaxin (which peaks around 12 weeks of gestation, remains stable during pregnancy, and declines postpartum)—are also theorized to contribute to changes in joint stability.[148] It is postulated that hormones may change the elasticity of ligaments by affecting collagen metabolism, which then increases laxity of the joints during pregnancy. Supporting this, Kristiansson et al. demonstrated that serum concentrations of a collagen turnover marker were significantly correlated with pelvic pain in pregnancy.[101] The concept of hormonal changes impacting joint stability is controversial and has been disputed in the literature. A 2012 systematic review especially called the effects of relaxin into question because of a lack of strong evidence in the literature.[7] Nevertheless, the combination of biomechanical and hormonal changes has an impact on the increased demand placed on the musculoskeletal system in pregnancy and postpartum, thereby creating the potential for musculoskeletal pain.

Musculoskeletal pain during pregnancy can arise from numerous areas, including the lumbar spine, pelvic girdle, hips, and PFMs. It is essential to identify musculoskeletal conditions that may cause severe, acute pain during pregnancy and delivery, such as sacral insufficiency fractures, fractures associated with transient osteoporosis of pregnancy, coccygeal fractures or luxation, and pubic symphysis separation.[93] It is also important to remember that musculoskeletal pain etiologies that occur in the nonpregnant state can also occur during pregnancy.

PGP is the most common cause of back and pelvic pain in pregnancy.[186] PGP is defined as pain that occurs between the posterior iliac crest and the gluteal fold, particularly in the region of the SIJs. PGP can radiate to the posterior thigh and may include pubic symphysis pain, which can radiate to the anterior thigh.[29] Of note, while low back, buttock, and leg pain during pregnancy are often assumed to be “sciatica” by patients and providers, the incidence of true lumbosacral radiculopathy, plexopathy, or sciatic neuropathy in pregnancy has been found to be low;[29] thus PGP should be considered the most likely diagnosis when such pain patterns are seen without clear neurologic deficits.

PGP is theorized to arise from changes in pelvic joint stability. This, in turn, is affected by pregnancy-related hormonal fluctuations and biomechanical alterations in form closure of the bony pelvis and force closure of the myofascial soft tissues as well as disruptions in motor control related to changes in the center of gravity.[184] No relationship has been shown between increased SIJ laxity and pelvic pain during pregnancy; however, there was a correlation between asymmetric laxity of the SIJs and pain during pregnancy.[55] Females with PGP typically demonstrate diminished endurance capacity for sitting, walking, and standing, but they also have particular difficulty with transitional and asymmetric activities (e.g., turning in bed, getting in and out of a car, and raising one leg to put on pants).[29] To diagnose PGP, lumbar causes of pain must be excluded, and the pain must be reproducible by specific clinical tests (see later).

Epidemiology

PGP is the most common musculoskeletal complaint in pregnancy, affecting 20% to 65% of pregnant females[68] and 8% to 20% of postpartum females at up to 3 years after delivery.[75,192] PGP prevalence may be underrepresented because of underreporting and underdiagnosis. The majority of PGP cases resolve by 12 weeks postpartum, but long-term symptoms persist in some patients.[63,159] Risk factors for worsened PGP during pregnancy and postpartum include a prepregnancy history of low back pain, prior pelvic trauma, parity, and strenuous workload.[29,63,186] Persistent PGP postpartum has been found to be more prevalent after cesarean delivery compared with vaginal delivery.[26,129] Breastfeeding may provide a small protective effect against persistent PGP,[24] while emotional distress during pregnancy has been associated with increased PGP risk.[25] PFM pain has been identified in 70% of females with pregnancy-related PGP in the second trimester[68] and in 52% of postpartum females with chronic lumbopelvic pain that began during pregnancy.[62]

Diagnosis and Physical Examination

History and physical examination are the primary diagnostic tools for evaluation of the pregnant or postpartum patient presenting with low back or pelvic pain. In addition to obtaining a detailed pain history as described earlier, including assessment of psychosocial factors, one should pay careful attention to any prior history of low back pain or pelvic trauma. Radicular lower extremity pain, paresthesia, weakness, groin pain, difficulty with weight bearing, and new-onset bowel or bladder symptoms are red flags that may signify potential nerve impingement, cauda equina syndrome, or fracture. Any of these symptoms warrant seeking emergent care.

Physical examination should include a neurologic examination of the lower extremities and a musculoskeletal examination of the pertinent areas of the patient's complaints, focusing on the spine, abdomen, pelvis, and hips. With regard to PGP, Vleeming et al.[186] proposed validated maneuvers for the diagnosis of pregnancy-related PGP, including pain provocation tests (posterior pelvic pain provocation [P4]/thigh thrust test, Patrick/FABER [*f*lexion, *ab*duction, *e*xternal *r*otation] test, Gaenslen test, modified Trendelenburg test) and pain palpation tests (pubic symphysis palpation, long dorsal ligament palpation). The P4 and FABER tests are the most sensitive tests for pregnancy-related PGP.[186] Functional stability testing should also be performed with the active straight-leg raise test, which can help predict response to stabilization exercises in PT if the patient has improvement in symptoms with external manual compression of the pelvic ring.[29,67,186]

Treatment

The mainstay of treatment for PGP in pregnancy and postpartum is individualized PT that includes mechanical assessment, postural correction, and stabilizing exercises.[159,186] The PT exercise program typically focuses on pelvic stabilization exercises to enhance motor control and muscular force closure of the pelvis.[67,159,186] The addition of PT exercises promoting muscular strengthening of the entire spine is also recommended.[25] Pelvic stabilization belts are frequently used for symptomatic treatment because they provide passive external force closure of the pelvis. SIJ belts, in particular, are a safe and effective treatment option for PGP in pregnancy, leading to improved pain and function over time.[67]

PFPT can be performed with clearance from the obstetrician if the patient has PFM pain or dysfunction. Pelvic manipulations can be beneficial for symptomatic relief, though it is unclear if they produce structural change.[126,159] A recent systematic review assessing eight RCTs on osteopathic manipulation treatment (OMT) of low back and PGP during and after pregnancy demonstrated significant moderate effects on pain reduction and functional improvements, although results were limited by small sample sizes, variable study protocols, and lack of long-term follow up.[71] OMT can be considered as an adjunctive treatment for symptomatic management of PGP in pregnancy and postpartum. Acupuncture is another complementary treatment for lumbopelvic pain that was found to be safe and effective in pregnancy in a recent systematic review, though further high-quality studies are needed. Although there is no evidence to specifically support massage, specific pillow use, or education (apart from PT treatment) in the treatment of PGP, these are clinically often found to be beneficial. Ice and acetaminophen are also often used in this patient population. In general, bedrest should especially be avoided because of the risks for venous thromboembolism, bone demineralization, and deconditioning.[4]

Acetaminophen is a first-line analgesic in pregnancy and is generally considered safe, although recent studies have demonstrated a correlation with increased risk of childhood asthma.[17] There may also be an association between in utero exposure to acetaminophen and development of attention-deficit/hyperactivity disorder or behavioral difficulties.[164,169] NSAIDs are not recommended during pregnancy, particularly in the third trimester, because of risk for premature ductus arteriosus closure and adverse fetal events, but they can safely be resumed in the postpartum period even when breastfeeding. Lidocaine patches for topical pain management and cyclobenzaprine as an antispasmodic/muscle relaxant are both relatively safe in pregnancy (pregnancy class B*) and can be used for symptomatic relief. Low-dose opioids are generally considered safe and may be necessary when severe pain does not respond to other rehabilitation treatments.[52] Injectable medications, including steroids (pregnancy category C*), can be considered on a case-by-case basis for severe recalcitrant pain.

Chronic Pelvic Pain

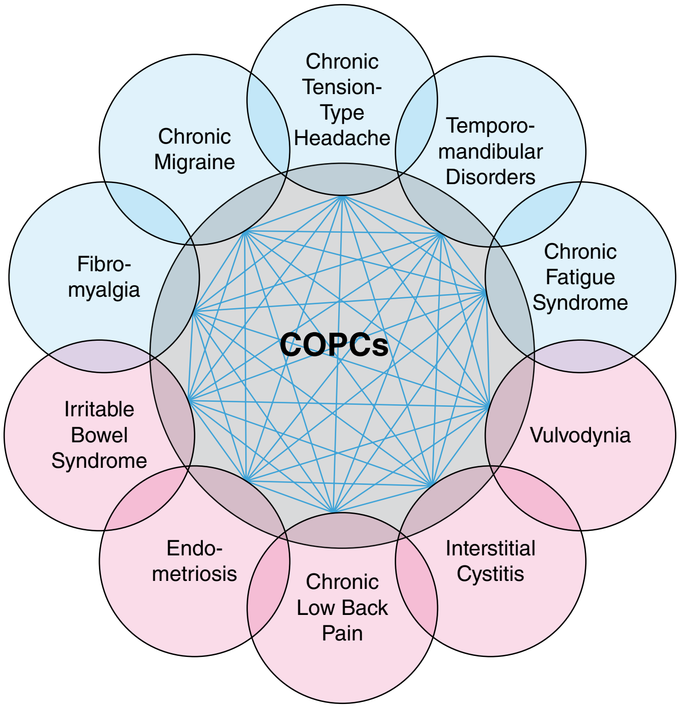
Definitions and Etiology

CPP—defined as noncyclic pain of 6 months' duration or more that localizes to the anatomic pelvis, anterior abdominal wall at or below the umbilicus, to the lumbosacral back or the buttocks, leading to disability,[22] has been reported to affect anywhere from 6% to 25% of females worldwide.[14] CPP is a broad term encompassing a multitude of potential etiologies and associated diagnoses (see Table 39.1). It is important to recognize that CPP differs significantly from acute pelvic pain in that the severity of CPP need not correlate with degree or presence of physical pathology, and CPP does not respond well to standard medical and surgical treatments.[40] CPP places a large burden on individual patients and the health care system; these patients undergo significantly more emergency department visits, imaging tests, and hysterectomies than those without CPP, underscoring the importance of appropriate diagnosis and treatment of these patients.[78]

Chronic Overlapping Pain Syndromes

CPP was previously studied in a compartmentalized fashion, considering the pathologies of each organ system independently. Recently, it has increasingly become clear that CPP is multifactorial and nonspecific, often with numerous overlapping etiologies and overlying psychiatric comorbidities. Viewed in this light, CPP shares obvious similarities with several chronic nonpelvic pain conditions, all of which share the common underlying central mechanisms, and has been shown to be frequently comorbid with these conditions.[102] These highly coprevalent chronic pain conditions have recently come to be recognized by the National Institutes of Health and the US Congress as chronic overlapping pain conditions (COPCs), of

*Of note, in response to concerns about the confusing nature of the prior US Food and Drug Administration pregnancy categories (A, B, C, D, X), a new descriptive labeling system was introduced, known as the Pregnancy and Lactation Rating Rule (PLRR), which became effective in 2015, but the pregnancy categories are included here as they are still frequently referenced in practice.[139,188]



• **Fig. 39.6** Chronic pelvic pain and the chronic overlapping pain conditions *(COPCs)*. Diagnoses in magenta circles can be seen in chronic pelvic pain. (Adapted from Kumar R, Scott K. Chronic pelvic pain and the chronic overlapping pain conditions in women. *Curr Phys Med Rehabil Rep*. 2020;8[3]:207-216.)

which there are 10 currently identified (Fig. 39.6).[181] As highlighted in the figure, 5 of the 10 conditions are localized within the pelvis and thus can be categorized as CPP: IBS, vulvodynia, interstitial cystitis/bladder pain syndrome (IC/BPS), endometriosis, and chronic low back pain.[118] We delve further into four of these conditions in the following sections (refer to Chapter 34 for further detail on the COPC of chronic low back pain).

Central sensitization is felt to be the common underlying mechanism shared by the COPCs and is defined as an amplification of neural signaling within the central nervous system that elicits pain hypersensitivity.[131] Central sensitization can be quantified using the Central Sensitization Inventory (CSI), a validated questionnaire.[130] There are also several maladaptive pain behaviors that are seen commonly in the COPCs, including pain catastrophizing, kinesiophobia, and low self-efficacy, which correlate highly with disability (refer to Chapter 38 for more details). Identifying these psychometric parameters in patients with CPP is integral to providing effective counseling and tailored treatment planning.[102,180]

Interstitial Cystitis/Bladder Pain Syndrome

The current agreed upon definition for IC/BPS comes from a consensus statement issued by the Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction, which defines the condition as "an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the urinary bladder, associated with LUTS of more than six weeks duration, in the absence of infection or other identifiable cause."[82,83,123] In the current understanding, IC and BPS are often used interchangeably; although IC had previously been reserved for those cases in which specific cystoscopic findings were seen, such as glomerulations on hydrodistention, more recently this distinction has been abandoned because the presence of these findings is variable and inconsistent with clinical symptoms.[123,137] IC/BPS is commonly diagnosed in the fourth decade or after and has been reported as being more prevalent in females (though this may be because of an underestimation of the prevalence in males).[83] Typical symptoms include bladder or pelvic pain, pressure, or discomfort; constant urge to void; and urinary frequency.[47] The current mechanistic understanding of IC/BPS remains elusive. Proposed etiologies have included genetic predisposition, neurogenic inflammation, epithelial dysfunction, autoimmune, infectious, toxic urinary components, psychosomatic mechanisms, and peripheral/central neurologic sensitization (see prior section on COPCs).[5,59] IC/BPS has been associated with PFMP; however, it is unclear whether this is causative or a secondary phenomenon.[83] Diagnosis of IC/BPS is made primarily by careful history, physical examination, ruling out confusable disorders such as urinary tract infection, nephrolithiasis, and underlying malignancy. Cystoscopy and urodynamic studies are not necessary for diagnosis, but they can be used to exclude other suspected etiologies. Multimodal management is recommended, including pharmacologic, behavioral, and PFPT treatments.[47] Comanagement with urology or urogynecology is recommended.

Endometriosis

Endometriosis is defined as the presence of endometrial glands and stroma outside the endometrial cavity and uterine musculature. As noted previously, central sensitization is felt to be an underlying mechanism in the pain associated with endometriosis (see prior section on COPCs). The CSI has been utilized to determine that the subpopulation of patients with endometriosis who have deep dyspareunia and bladder and/or pelvic floor tenderness exhibit a high degree of central sensitization.[136] Symptoms of endometriosis may include disabling menstrual cramps, CPP, low back pain, pain during or after intercourse, painful bowel movements, painful urination during menstrual periods, heavy menstrual periods, premenstrual spotting or bleeding between periods, and/or a history of infertility. Laparoscopic excision of the implants with histologic confirmation is considered the diagnostic standard of care. Females with endometriosis should be evaluated for cooccurring high-tone pelvic floor dysfunction. Comanagement with gynecology is recommended for surgical and pharmacologic treatment options.

Irritable Bowel Syndrome

IBS is a disorder characterized by altered bowel function in terms of frequency and/or consistency (constipation, diarrhea, or mixed) with associated abdominal pain and often bloating or distention, with symptoms present for at least 6 months.[35,105] Estimates of the prevalence of IBS range from 7% to 16% in the United States, more common in females and younger individuals.[35] IBS is classified as a disorder of gut-brain interaction. Although it was previously thought that there was no structural or biochemical abnormality underlying IBS, more recent research has identified both central and peripheral gut-specific pathophysiologic mechanisms.[36,91] Potential mechanisms contributing to IBS include abnormal motility, heightened visceral perception, pelvic floor dysfunction, psychologic distress, and intraluminal factors irritating the bowel, such as lactose, other sugars, and bile acid.[38]

Symptoms suggestive of IBS are present in approximately 35% to 50% of patients with CPP.[121] Patients with IBS will typically have a normal general physical examination but may have an abnormal pelvic floor examination. Diagnostic studies should include a complete blood count and a C-reactive protein or fecal calprotectin. Treatment ranges from education and lifestyle changes (exercise, stress reduction, and sleep regulation) to regulating diet to optimizing the use of nonirritant oral agents (e.g., bulking agents, natural laxatives). Referral to gastroenterology is appropriate to ensure thorough screening and consider initiation of IBS-specific pharmacotherapy.[35,105]

Vulvar Pain and Vulvodynia

Vulvar pain is a common gynecologic complaint, affecting 8% to 15% of females of reproductive age.[11,85,147] Vulvar pain is divided into two groups: vulvar pain caused by a specific disorder and vulvodynia. Vulvar pain resulting from a specific disorder may be infectious, inflammatory, neoplastic, neurologic, traumatic, iatrogenic, or hormonal in origin. Vulvodynia is defined as vulvar pain that has been present for at least 3 months without a clearly identifiable cause.[30] It is described by location, provocation, onset, and temporal patterns. Potential factors associated with vulvodynia include comorbid pain syndromes, genetics, hormonal factors, inflammation, musculoskeletal issues, neuroproliferation, psychosocial factors, and structural defects.[30] The main physical finding is a positive Q-tip test suggestive of vulvar allodynia.[135] Patients with vulvodynia also benefit from a thorough pelvic floor evaluation, as 90% of females with provoked vestibulodynia also have high-tone pelvic floor dysfunction.[84,150] A multidisciplinary approach for treatment of vulvodynia is ideal and may include PFPT, medical therapy, and psychotherapy.[182] There is lack of evidence to support medication treatment options for vulvodynia. Often in clinical practice, oral neuromodulators are used for the treatment of generalized vulvodynia, and compounded topical therapies are used for provoked vestibulodynia. Vestibulectomy or surgical excision can be effective for the treatment of provoked vestibulodynia when patients have not responded to medical treatments, but either should be considered as a last resort.

Treatment of Chronic Pelvic Pain

As alluded to earlier, CPP does not respond well to standard treatment interventions because it stems from altered pain perception at the level of the central nervous system. Thus successful treatment of chronic pelvic pain necessitates a holistic treatment paradigm rather than a trial-and-error approach with a series of narrowly focused procedures that would be better suited to address acute, localized pain. The treatment approach for CPP should take into account biopsychosocial factors contributing to pelvic pain, such as comorbid mood disorders and COPCs, history of abuse and trauma, central sensitization, and maladaptive pain coping strategies. We feel it is essential for providers to use this big picture understanding of the patient's background to educate them on concepts of pain biology and central sensitization in a tailored and approachable way because improving patients' understanding of their condition allows for the setting of realistic expectations and helps foster a therapeutic alliance. In particular, it is crucial for providers to avoid suggesting to the patient that in light of underlying central mechanisms and comorbid psychiatric conditions, their pain is not "real." This characterization is both false and harmful, and it undermines efforts to help these patients by reducing their sense of agency and access to care and support.[88]

It is fundamental to address lifestyle factors, advising patients on optimizing sleep, diet, and cardiovascular exercise. Psychiatric comorbidities must be addressed, and pain-focused cognitive behavioral therapy and motivational interviewing can also be very beneficial. There is emerging evidence showing the benefits of mindfulness-based practices (e.g., yoga, tai chi, qigong) in CPP, although further research into these alternative treatments is needed.

From the pharmacologic standpoint, neuromodulatory agents are effective in CPP, as they can address pain, central sensitization, anxiety, and sleep.[28,102] TCAs (e.g., nortriptyline, amitriptyline) and related medications, such as trazodone and cyclobenzaprine, may be used to address pain, mood, and sleep disorders associated with myofascial pain syndromes, but they can cause anticholinergic side effects such as dry mouth, constipation, or urinary retention. Moreover, emerging evidence demonstrates a potential for both short- and long-term cognitive side effects, including dementia.[77,156] If there is a neurogenic or central sensitization component, antiepileptics (e.g., gabapentin or pregabalin) or SNRIs (e.g., duloxetine or venlafaxine) may also be useful. SNRIs may be better tolerated than antiepileptics because they have less sedating side effects.[28]

More invasive interventions should be undertaken with caution in chronic pelvic pain because localized procedures are less likely to be effective than in other patient populations. These interventions may include peripheral nerve or superior or inferior hypogastric plexus blocks, as well as peripheral nerve, sacral nerve root, or spinal cord stimulators. There is a limited role for surgery in the management of chronic pelvic pain.[102]

Summary

In summary, pelvic floor disorders greatly affect the quality of life of patients of both sexes, including pregnant and postpartum patients. Diagnosis of pelvic floor disorders can be made with a thorough history and physical examination. It is important to include musculoskeletal and neurologic causes in the differential for urinary and defecatory dysfunction and incontinence and pelvic pain, to provide appropriate rehabilitation treatments in a timely manner and avoid unnecessary surgeries or invasive procedures. Standard of care treatment includes a trial of conservative therapies, including PFPT, lifestyle/behavioral modifications, and medications, which can often provide significant symptom reduction and an improved quality of life.

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